

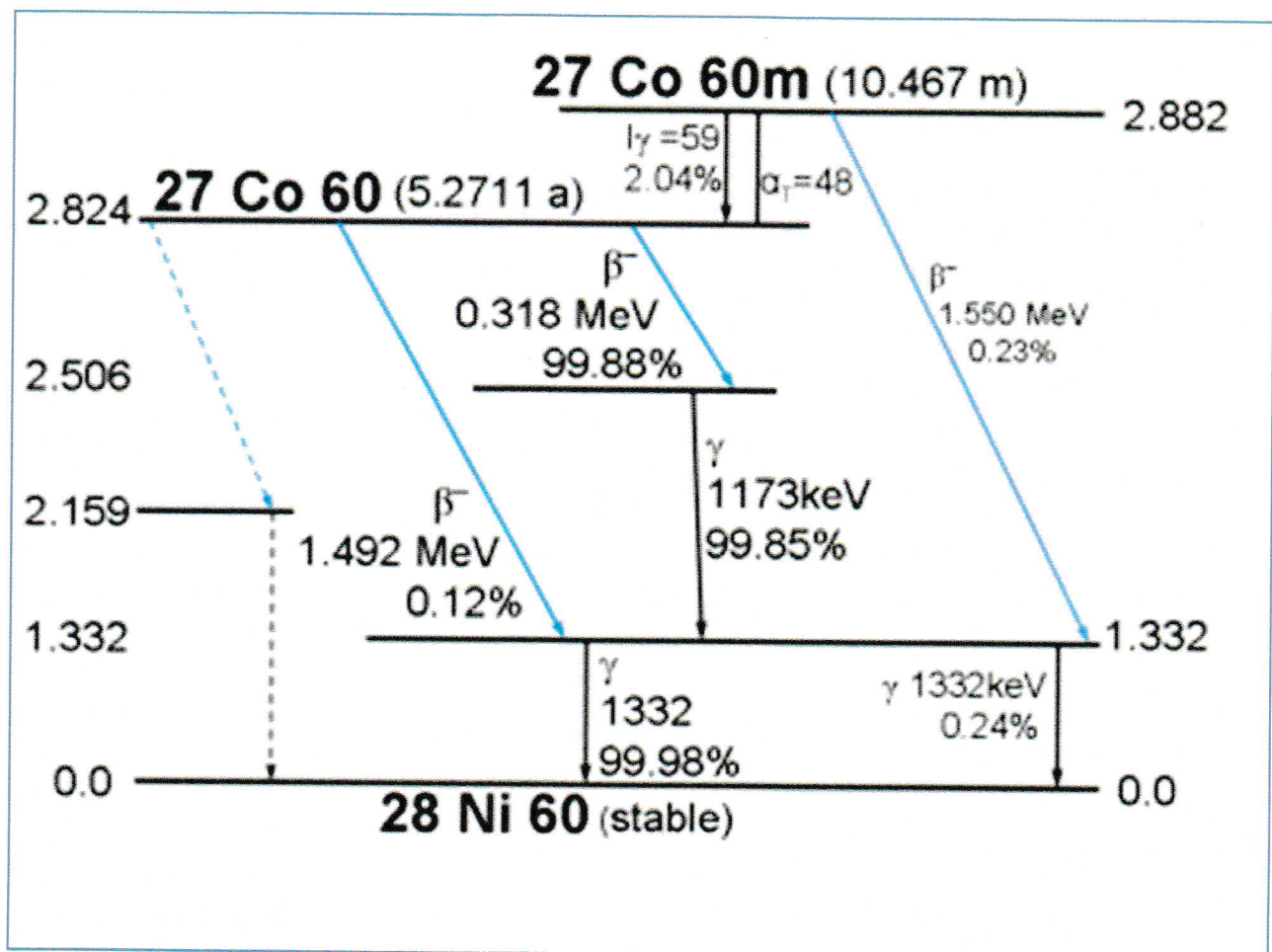
Cobalt Activation and Decay

Isotopic Constants for ^{59}Co , ^{60}Co and $^{60\text{m}}\text{Co}$

(reference chart key!!)

Co59 $7/2^-$	Co60 2^+	Co60 5^+
100	10.47 m	5.271 a
	IT 58.6, e^-	β^- .318, ...
	β^- 1.6 $\omega, \dots$	γ 1332.5, 1173.2, ...
σ_γ (21+16), (39+35)	γ 1332.5 $\omega, \dots$	σ_γ 2.0, 4 E 2.824
58.933200	σ_γ 6E1, 2.3E2	

Decay Levels for ^{60}Co (ground state) and $^{60\text{m}}\text{Co}$ (isomeric state)



Uranium Series ($4n + 2$)*

Nuclide	Historical name	Half-life	Major radiation energies (MeV) and intensities†		
			α	β	γ
$^{238}_{92}\text{U}$	Uranium I	$4.51 \times 10^9 \text{ y.}$	4.15 (25%) 4.20 (75%)	---	---
$^{234}_{90}\text{Th}$	Uranium X_1	24.1d	---	0.103 (21%) 0.193 (79%)	0.063c‡ (3.5%) 0.093c (4%)
$^{234}_{91}\text{Pa}^m$	Uranium X_2	1.17m	---	2.29 (98%)	0.765 (0.30%) 1.001 (0.60%)
99.87% 0.13% $^{234}_{91}\text{Pa}$	Uranium Z	6.75h	---	0.53 (66%) 1.13 (13%)	0.100 (50%) 0.70 (24%) 0.90 (70%)
$^{234}_{92}\text{U}$	Uranium II	$2.47 \times 10^5 \text{ y}$	4.72 (28%) 4.77 (72%)	---	0.053 (0.2%)
$^{230}_{90}\text{Th}$	Ionium	$8.0 \times 10^4 \text{ y}$	4.62 (24%) 4.68 (76%)	---	0.068 (0.6%) 0.142 (0.07%)
$^{226}_{88}\text{Ra}$	Radium	1602y	4.60 (6%) 4.78 (95%)	---	0.186 (4%)
$^{222}_{86}\text{Rn}$	Emanation Radon (Rn)	3.823d	5.49 (100%)	---	0.510 (0.07%)
$^{218}_{84}\text{Po}$	Radium A	3.05m	6.00 (~100%)	0.33 (~0.019%)	---
99.98% 0.02% $^{214}_{82}\text{Pb}$	Radium B	26.8m	---	0.65 (50%) 0.71 (40%) 0.98 (6%)	0.295 (19%) 0.352 (36%)
$^{218}_{85}\text{At}$	Astatine	~2s	6.65 (6%) 6.70 (94%)	? (~0.1%)	---
$^{214}_{83}\text{Bi}$	Radium C	19.7m	5.45 (0.012%) 5.51 (0.008%)	1.0 (23%) 1.51 (40%) 3.26 (19%)	0.609 (47%) 1.120 (17%) 1.764 (17%)
99.98% 0.02% $^{214}_{84}\text{Po}$	Radium C'	164 μ s	7.69 (100%)	---	0.799 (0.014%)
$^{210}_{81}\text{Tl}$	Radium C''	1.3m	---	1.3 (25%) 1.9 (56%) 2.3 (19%)	0.296 (80%) 0.795 (100%) 1.31 (21%)
$^{210}_{82}\text{Pb}$	Radium D	21y	3.72 (.000002%)	0.016 (85%) 0.061 (15%)	0.047 (4%)
$^{210}_{83}\text{Bi}$	Radium E	5.01d	4.65 (.00007%) 4.69 (.00005%)	1.161 (~100%)	---
~100% .00013% $^{210}_{84}\text{Po}$	Radium F	138.4d	5.305 (100%)	---	0.803 (0.0011%)
$^{206}_{81}\text{Tl}$	Radium E''	4.19m	---	1.571 (100%)	---
$^{206}_{82}\text{Pb}$	Radium G	Stable	---	---	---

*This expression describes the mass number of any member in this series, where n is an integer.

Example: $^{206}_{82}\text{Pb}$ ($4n + 2$)..... $4(51) + 2 = 206$

†Intensities refer to percentage of disintegrations of the nuclide itself, not to original parent of series.

‡Complex energy peak which would be incompletely resolved by instruments of moderately low resolving power such as scintillators.

Data taken from: Table of Isotopes and USNRC-TR-862